BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Supplementary Material

A. Exact labels and admissible bounds

The archived predecessor audit evaluates every stored nonnegative state–goal label. Let R be occupied blocker closure for required removals and insertions. Without capacity or inspection obligations, the formula $|A \triangle G| + 2|R \cap A \cap G|$ agrees with all 1,802,728 audited labels. This is a finite-census equality, not a general theorem about arbitrary assembly domains or visual perception.

For BrickPlan's lower bound, initially mismatched sites each require an operation. Every occupied target site in the required removal closure must be detached and reattached. Every unfinished job needs one audit. These counts refer to disjoint required operations. Recursively including occupied path blockers can only add necessary removals. Ignoring capacity scheduling and some connectivity effects relaxes the problem, so $h(x) \leq C^*(x)$. The verifier checks this inequality at every oracle node.

For every nonterminal product node, the certificate checks $C^*(x) = \min_{a \in A(x)}[1 + C^*(T(x, a))]$, with terminal cost zero. Physical edges are compared with independent runtime transition enumeration and inverse execution; audit edges are replayed as well. Every retained product state reaches a terminal. The physical graph is reversible within its capacity-constrained component. Audits add a persistent bit without removing a physical action, and all admitted jobs can be exposed. Consequently, legal successors do not become genuine irreversible dead ends. Budget exhaustion and extra cost remain possible.

B. Construction details

The machine-readable source plan distinguishes development and new-source groups before new-source solver discovery. Legacy tasks retain their earlier selection bias. Additional task generation uses deterministic seeds, sampled states and work orders, and solvability admission. Geometric pruning was refined after an initial collision-rejecting pass, before new-source policy results. Both passes and their exact retained source indices are archived. No difficulty-based rejection is used for these new-source tasks.

Four geometry pairs share registered pieces and original transforms, start, goal, capacity, jobs, and proposed action sequence. Induced fixed contexts are separately qualified. All four answer-changing pairs come from the roadster development source and have unequal fixed-part counts. Three equal-count/type context candidates yield zero retained answer-changing pairs; the full census is in data/geometry-probe-audit.json. Original/grayscale pairs preserve all geometry and labels.

The 72 interior-order pairs are balanced over conflict bins 2–5, 6–10, and 11+. Legal and reordered plans share exact prefixes up to the first conflict, endpoints, and action multisets. Legal witnesses have no repeated states. The stored audit exposes each conflict position, so readers can run other prefix controls.

Table 1. A prefix-only software control returns the first conflict if it appears within k executed operations, otherwise "all legal." The denominator is all 144 continuation questions.

k	0	2	3	5	8	11	32
Correct	72	80	87	96	115	122	144

Table 2. Measured finite-horizon choice sets for all 230 next-action questions. Any/all refer to optimistic/pessimistic ties; uniform tie correctness is an exact average, not a sampled policy result.

Depth	Choices	All optimal	Any optimal	Uniform tie
1	230	172	230	87.6%
2	230	186	230	89.9%
4	230	178	230	87.2%
8	230	197	230	92.2%

C. Shallow choice and control diagnostics

The depth-one minimizing set contains an optimal action at all 230 selected checkpoints. Therefore the diagnostic bank does not demonstrate that shallow reasoning necessarily excludes the correct answer. Deeper horizons can change tie composition nonmonotonically. The full table is retained without redefining difficulty to match its results.

All 4,060 raw policy task-run records are archived in data/software-control-records.json.gz. They include one oracle seed and three seeds for nine other policies, with actions, outcomes, expansions, timings, and regrets. The release checker recomputes every overall,

Table 3. Structural policy completion by operation-length bin. Each nonoracle policy has three tie-order seeds per episode.

Control	D1	D2	D3	D4
Closure greedy	48/48	134/141	136/138	66/108
Capacity dispatch	48/48	135/141	129/138	55/108
Closure A* (100)	48/48	140/141	135/138	39/108
Closure A* (1,000)	48/48	141/141	135/138	80/108
Closure A* (full)	48/48	141/141	138/138	108/108
Hamming A* (full)	48/48	141/141	138/138	108/108

tier, source, and seed aggregate. The full search cap of 500,000 is a resource ceiling, not actual expansion usage. Oracle loading and cached local timing are not comparable to calibrated online search.

D. Recovery, diagnosis, and censoring

Traverse only edges with $C^*(T(x,a)) = C^*(x) - 1$ from pre-error state b. The reached states are exactly the union of optimal completions: each traversal is an optimal prefix, and every optimal prefix satisfies that equality. Remove the terminal state to obtain the repair target $\mathcal{C}_b^-$. The generator's delayed checkpoints remain outside this set after an error and lag 1, 3, or 5. Independent forward BFS verifies their shortest return distances, which range from 1 to 6. The 333 questions comprise 117, 109, and 107 checkpoints at those lags.

Each positive-excess legal action creates its own spontaneous event; later actions update every active event. A second error never replaces the first. Immediate recovery and lag-specific delayed cohorts are reported separately. Unfinished events are censored; terminal events that never restore a nonterminal corridor state are observed failures. Reaching a goal after a costly detour earns completion credit, but not automatic corridor-repair credit.

An accepted diagnosis concerns legal positive-excess events since the last diagnosis or episode start. The first later diagnosis is attached to each previously undiagnosed event, with its sign and lag. True positive diagnosis checkpoints open replanning-suffix records. Always-negative or selectively requested diagnoses can yield misleading accuracy, so report event detection and query coverage alongside it.

Plan-survival reports retain the first cost loss even after recovery, and include explicit eligible/observed/unknown counts per operation index. They are descriptive task-budget curves, not Kaplan–Meier population estimates. Conditions should use matched assignments and registered source strata. Missing episodes must not disappear from denominators.

D.1. Regret conservation

For a successful trace, accepted attaches A_c and detaches D_c satisfy $A_c - D_c = \Delta$, the target-minus-start occupancy count. There are exactly u accepted previously unfinished audits. Duplicate audits are rejected. If N_{invalid} is the number of charged invalid attempts, then

$$T = 2D_c + \Delta + u + N_{\text{invalid}}, \tag{1}$$

$$C^*(x_0) = 2D^* + \Delta + u, \tag{2}$$

$$T - C^*(x_0) = 2(D_c - D^*) + N_{\text{invalid}}. \tag{3}$$

This audit-inclusive identity reconciles detachment regret and total excess. Rejected actions consume budget; stale HTTP revisions and malformed request envelopes are rejected before mutation and are not action events.

E. Visual stimuli and actual model calls

All executable geometry uses original source poses and LDraw triangle meshes. The 153 distinct part meshes contain 212,778 triangles. ASSET-LICENSES.json preserves source models, authors, URLs, and part-level notices; attribution and licenses remain separate from code. Qualification proxies do not replace the displayed shapes. Figure provenance records exact states, seeds, bank hash, and image hashes.

The public renderer supersamples by two before Lanczos downsampling to native HD. Display canvases account for device-pixel ratio, and native zoom avoids CSS-only enlargement of low-resolution pixels. Catalog tiles are 480×360 and catalogs are 1920 pixels wide. Models receive image content, not text describing images. Preprocessing is separately declared: the local SmolVLM case resizes to longest edge 512, with splitting disabled. HD display does not imply native-HD model processing.

The first local case contains four actual inference calls on public E069, seed 0. All return descriptive prose instead of an executable JSON action. The executor rejects each, and the controller stops at its four-call limit. Every prompt, raw response, image hash, latency, measured token count, action, and feedback is archived. No oracle action or fabricated explanation replaces model output. This is a failed integration example. It supplies neither an estimate of SmolVLM accuracy nor evidence about larger VLMs.

F. Protocol and reproducibility

Study runs bind participant, condition, seed, assignments, and bank hash. The evaluator replays every mutation, recomputes outcomes, validates view

Table 4. Metric denominators and treatment of missing outcomes.

Metric	Eligible denominator	Missing / termination rule
Completion / optimal completion	All assigned episodes	Unstarted and unfinished are failures
Successful mean regret	Successfully finished episodes	Null for failures; report success coverage
Avoidable-cost choice	Legal decisions with zero- and positive-excess alternatives	Invalid attempts excluded here, charged in total
Budget-loss choice	Viable checkpoints with viable and nonviable legal choices	Other budget losses remain in raw records
True-unreachability choice	No eligible choices in retained components	Undefined, not empirical evidence of zero risk
Recovery bounds	Every eligible legal positive-excess event	Lower bound restored/all; upper adds censored
Diagnosis accuracy	Accepted diagnosis queries	Unqueried intervals are not inferred correct
Joint pair correctness	Fully assigned pairs	Missing member prevents joint credit
Probe accuracy	All assigned questions of that family	Missing or invalid response gets no credit

transcripts, and normalizes candidate permutations. Identical request retries are idempotent across restart; changed content under a used ID or stale revision is rejected. Atomic persistence rolls back in-memory mutations on write failure.

Practice exposes rich catalogs, solution review after submission, and live free 3D inspection. Study mode blocks those routes and downloadable private evaluator artifacts. The portable public release includes an explicit evaluator directory for reproducibility; it must remain on the evaluator host during a blind study. A local process working-directory convention is not a security sandbox.

The practice model manager supports local MLX and OpenAI-compatible vision calls, run/step/stop, durable outputs, and bounded calls. Server credentials are not included in browser inputs or exports. Stop prevents subsequent actions after an in-flight request returns; it cannot undo provider billing. Unknown calls after restart are retained as interrupted, never silently retried.

The population study runner is distinct. Its reported-usage mode reserves declared maximum input/output tokens and cost before each call. Adapters must account for images, enforce limits, and attest usage. Incomplete usage, timeout, malformed responses, or limit violations stop automatic continuation. Unavailable-accounting mode marks tokens and currency unavailable. Adapter/provider accounting is a declared trust boundary, not a provider-enforced spending quota.

The main paper uses the unmodified official CVPR author-kit, pinned to commit 2917585. The retrieved kit identifies itself as 2026; this does not verify a future conference year's kit. The default PDF is a public research preprint. Review mode can be compiled separately after setting the actual submission year and paper ID. No custom margin or body-font compression is applied.

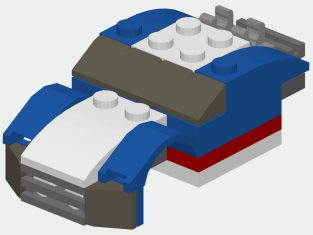
G. Source profiles

Profiles without episodes are zero-yield qualified geometry candidates, not additional benchmark tasks. Registered and movable counts describe different quantities; fixed geometry can dominate appearance.

Profile	Kit	Parts	Reg.	Move	Rack	Episodes	Max gap
Roadster body	31027	37	17	17	9	14	18
Kart deck	31027	25	18	18	5	14	10
Airframe	31028	24	21	21	8	14	4
Truck cab and roof	10156	76	18	18	12	10	6
House roof assembly	31009	221	18	17	5	14	4
Sports car upper body	5867	149	22	10	8	3	0
Source 10001 subassembly	10001	17	2	2	2	8	0
Source 10159 subassembly	10159	177	16	16	10	12	2
Source 10220 subassembly	10220	153	16	16	10	12	0
Source 10014 subassembly	10014	71	16	15	10	12	4
Source 10128 subassembly	10128	98	16	14	10	12	0
Source omr-31031 subassembly	31031	42	12	12	8	12	2
Airframe · support context 1	31028	23	21	—	8	0	—
Airframe · support context 2	31028	23	21	—	8	0	—
Roadster Body · support context 1	31027	37	17	—	9	4	4
Roadster Body · support context 2	31027	36	17	—	9	4	4
Sports Body · support context 1	5867	148	22	—	8	0	—
Sports Body · support context 2	5867	148	22	—	8	0	—
Truck Cab · support context 1	10156	75	18	—	12	0	—
Truck Cab · support context 2	10156	75	18	—	12	0	—

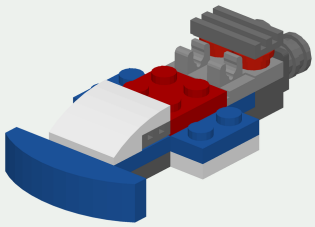
H. Original-source gallery

Roadster body | kit 31027



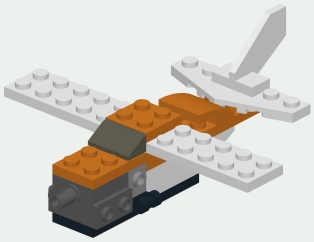
37 source parts; 17 registered
17 movable in the full-assembly component

Kart deck | kit 31027



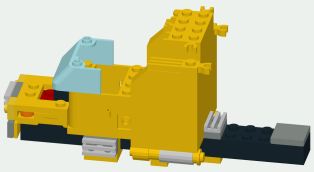
25 source parts; 18 registered
18 movable in the full-assembly component

Airframe | kit 31028



24 source parts; 21 registered
21 movable in the full-assembly component

Truck cab and roof | kit 10156



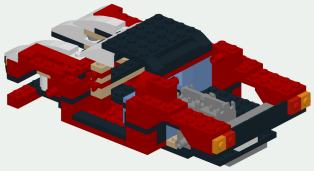
76 source parts; 18 registered
18 movable in the full-assembly component

House roof assembly | kit 31009



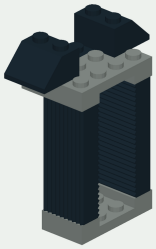
221 source parts; 18 registered
17 movable in the full-assembly component

Sports car upper body | kit 5867



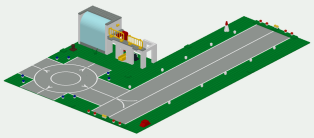
149 source parts; 22 registered
10 movable in the full-assembly component

Source 10001 subassembly | kit 10001



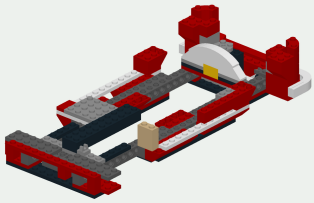
17 source parts; 2 registered
2 movable in the full-assembly component

Source 10159 subassembly | kit 10159



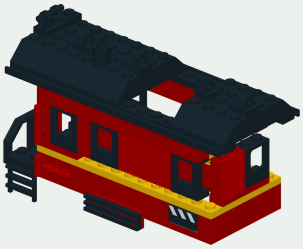
177 source parts; 16 registered
16 movable in the full-assembly component

Source 10220 subassembly | kit 10220



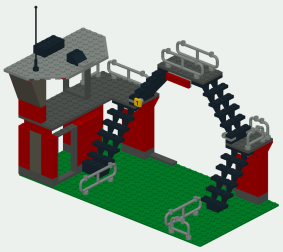
153 source parts; 16 registered
16 movable in the full-assembly component

Source 10014 subassembly | kit 10014



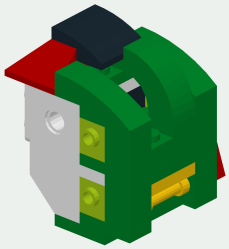
71 source parts; 16 registered
15 movable in the full-assembly component

Source 10128 subassembly | kit 10128



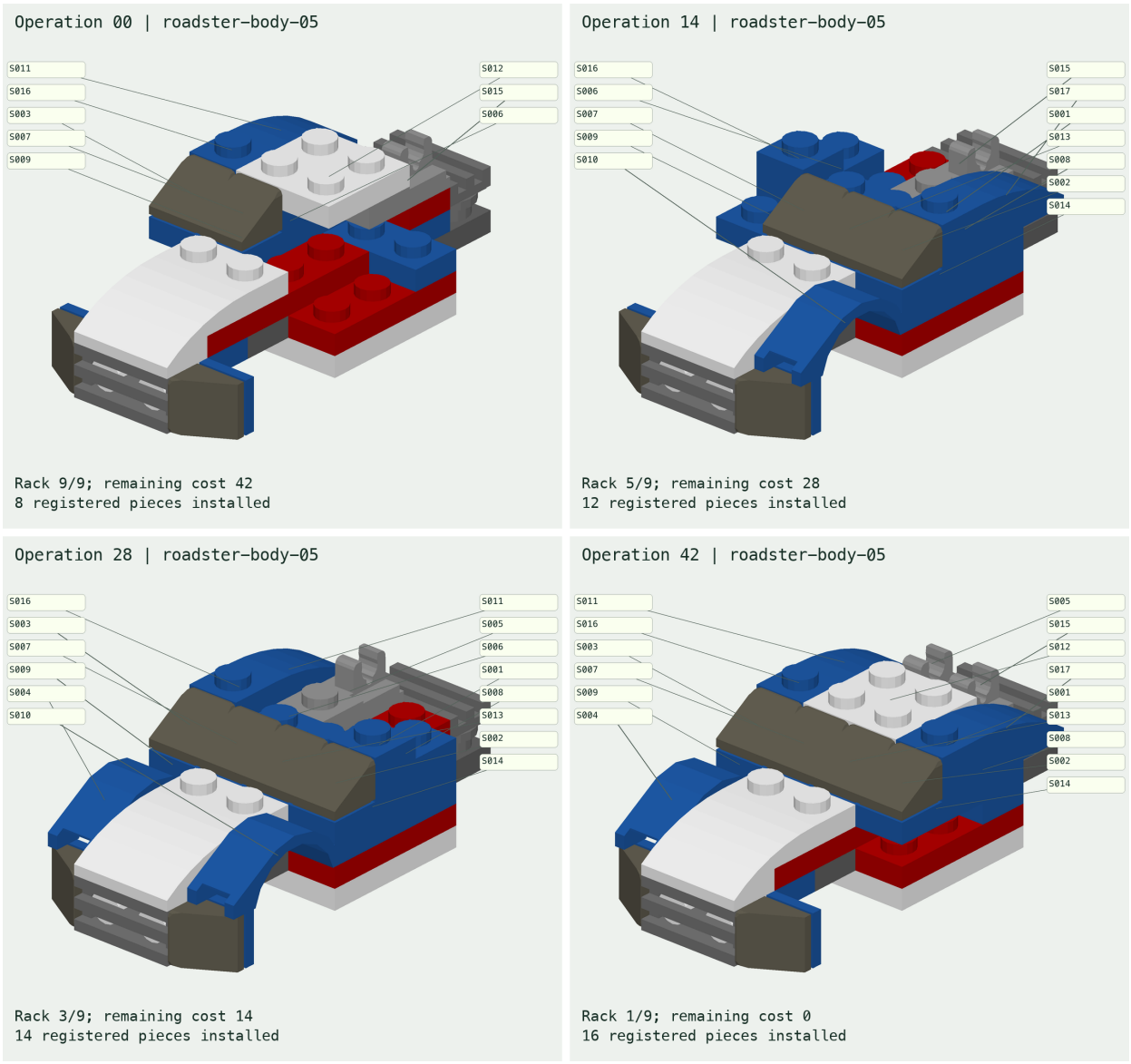
98 source parts; 16 registered
14 movable in the full-assembly component

Source omr-31031 subassembly | kit 31031



42 source parts; 12 registered
12 movable in the full-assembly component

I. A certified optimal resource replay



This is an exact software-oracle witness, not a VLM trajectory. All displayed states preserve source poses and certified connectivity. The figure provenance contains the task ID, seed, occupancy, ledger, step indices, and exact remaining cost. Rack holders and off-assembly work are explicit abstractions of the task.

J. Complete episode census

"Jobs" is the declared inspection count, "Done" its initially completed subset, and A* expansions use seed 0 with the closure bound. D1–D4 are length bins. Internal IDs are evaluator documentation; participant IDs are consistently permuted by seed. Exact starts, goals, jobs, witnesses, and oracle paths are available in the task bank.

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
roadster-body-01	D3	17	0	0	17	17	17
roadster-body-02	D4	17	0	0	32	32	32
roadster-body-03	D4	9	0	0	27	23	33
roadster-body-04	D4	9	0	0	27	23	38
roadster-body-05	D4	9	0	0	42	24	2555
roadster-body-06	D4	9	0	0	42	24	2555
roadster-body-07	D4	9	2	0	37	27	2185
roadster-body-08	D4	9	2	0	37	27	2139
roadster-body-09	D3	9	2	1	20	20	21
roadster-body-10	D3	9	2	1	18	18	18
roadster-body-11	D3	9	1	0	18	18	18
roadster-body-12	D3	9	1	0	18	18	18
roadster-body-13	D4	9	4	0	39	29	2131
roadster-body-14	D4	9	4	0	39	29	2129
kart-deck-01	D3	18	0	0	18	18	18
kart-deck-02	D4	18	0	0	25	25	25
kart-deck-03	D3	5	0	0	17	11	32
kart-deck-04	D3	5	0	0	17	11	32
kart-deck-05	D4	5	0	0	26	18	119
kart-deck-06	D3	5	0	0	24	16	90
kart-deck-07	D4	5	2	0	31	21	121
kart-deck-08	D4	5	2	0	29	21	218
kart-deck-09	D3	5	2	1	21	15	68
kart-deck-10	D3	5	2	1	23	19	42
kart-deck-11	D3	5	1	0	23	19	42
kart-deck-12	D3	5	1	0	21	15	68
kart-deck-13	D4	5	4	0	33	23	403
kart-deck-14	D4	5	4	0	33	23	392
airframe-01	D3	21	0	0	21	21	21
airframe-02	D4	21	0	0	33	33	33
airframe-03	D2	8	0	0	16	14	16
airframe-04	D2	8	0	0	16	14	16
airframe-05	D3	8	0	0	17	13	30
airframe-06	D3	8	0	0	17	13	30
airframe-07	D3	8	2	0	22	20	22
airframe-08	D3	8	2	0	22	20	22
airframe-09	D2	8	2	1	9	9	9
airframe-10	D2	8	2	1	9	9	9
airframe-11	D2	8	1	0	9	9	9
airframe-12	D2	8	1	0	9	9	9
airframe-13	D4	8	4	0	26	24	26
airframe-14	D4	8	4	0	26	24	26
truck-cab-01	D3	18	0	0	18	18	18
truck-cab-02	D4	18	0	0	27	27	27
truck-cab-03	D4	12	2	0	40	34	1522
truck-cab-04	D4	12	2	0	36	30	809
truck-cab-05	D3	12	2	1	21	19	22
truck-cab-06	D3	12	2	1	21	19	22
truck-cab-07	D3	12	1	0	21	19	22
truck-cab-08	D3	12	1	0	21	19	22
truck-cab-09	D4	12	4	0	42	36	1546

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
truck-cab-10	D4	12	4	0	38	32	831
house-roof-01	D3	18	0	0	18	18	18
house-roof-02	D3	18	0	0	22	22	22
house-roof-03	D3	5	0	0	21	17	37
house-roof-04	D3	5	0	0	21	17	37
house-roof-05	D3	5	0	0	22	18	45
house-roof-06	D3	5	0	0	22	18	45
house-roof-07	D4	5	2	0	27	23	54
house-roof-08	D4	5	2	0	27	23	50
house-roof-09	D3	5	2	1	17	17	19
house-roof-10	D3	5	2	1	17	17	23
house-roof-11	D3	5	1	0	17	17	23
house-roof-12	D3	5	1	0	17	17	19
house-roof-13	D4	5	4	0	33	29	76
house-roof-14	D4	5	4	0	31	27	68
sports-body-01	D2	22	0	0	10	10	10
sports-body-02	D2	22	0	0	11	11	11
sports-body-03	D3	8	2	0	17	17	17
source-10001-01	D1	2	0	0	2	2	2
source-10001-02	D1	2	0	0	1	1	1
source-10001-03	D1	2	2	0	5	5	5
source-10001-04	D1	2	2	0	6	6	6
source-10001-05	D1	2	0	0	2	2	2
source-10001-06	D1	2	0	0	1	1	1
source-10001-07	D1	2	2	0	4	4	4
source-10001-08	D1	2	2	0	4	4	4
source-10159-01	D2	10	0	0	11	11	11
source-10159-02	D2	10	0	0	9	9	9
source-10159-03	D2	10	2	0	12	12	12
source-10159-04	D2	10	2	0	11	11	11
source-10159-05	D3	10	4	0	21	21	21
source-10159-06	D2	10	4	0	14	14	14
source-10159-07	D2	10	0	0	10	10	10
source-10159-08	D2	10	0	0	14	12	45
source-10159-09	D2	10	2	0	14	14	14
source-10159-10	D2	10	2	0	9	9	9
source-10159-11	D2	10	4	0	12	12	12
source-10159-12	D2	10	4	0	16	16	16
source-10220-01	D1	10	0	0	8	8	8
source-10220-02	D1	10	0	0	7	7	7
source-10220-03	D2	10	2	0	14	14	14
source-10220-04	D2	10	2	0	13	13	13
source-10220-05	D2	10	4	0	14	14	14
source-10220-06	D3	10	4	0	19	19	19
source-10220-07	D1	10	0	0	7	7	7
source-10220-08	D2	10	0	0	11	11	11
source-10220-09	D2	10	2	0	12	12	12
source-10220-10	D2	10	2	0	10	10	10
source-10220-11	D2	10	4	0	14	14	14
source-10220-12	D2	10	4	0	14	14	14
source-10014-01	D2	10	0	0	10	10	10
source-10014-02	D3	10	0	0	21	21	22
source-10014-03	D2	10	2	0	10	10	10
source-10014-04	D3	10	2	0	20	20	20
source-10014-05	D4	10	4	0	26	26	29
source-10014-06	D3	10	4	0	21	21	21

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10014-07	D1	10	0	0	8	8	8
source-10014-08	D2	10	0	0	15	15	15
source-10014-09	D2	10	2	0	14	12	58
source-10014-10	D2	10	2	0	14	14	15
source-10014-11	D3	10	4	0	21	17	4372
source-10014-12	D2	10	4	0	15	13	52
source-10128-01	D1	10	0	0	5	5	5
source-10128-02	D2	10	0	0	14	14	14
source-10128-03	D2	10	2	0	15	15	15
source-10128-04	D3	10	2	0	18	18	18
source-10128-05	D3	10	4	0	24	24	24
source-10128-06	D3	10	4	0	18	18	18
source-10128-07	D2	10	0	0	10	10	10
source-10128-08	D1	10	0	0	8	8	8
source-10128-09	D2	10	2	0	11	11	11
source-10128-10	D2	10	2	0	14	14	22
source-10128-11	D2	10	4	0	15	15	15
source-10128-12	D3	10	4	0	19	19	19
source-omr-31031-01	D2	8	0	0	12	12	12
source-omr-31031-02	D1	8	0	0	6	6	6
source-omr-31031-03	D2	8	2	0	13	13	13
source-omr-31031-04	D2	8	2	0	11	11	11
source-omr-31031-05	D3	8	4	0	17	17	17
source-omr-31031-06	D3	8	4	0	20	20	20
source-omr-31031-07	D2	8	0	0	13	11	26
source-omr-31031-08	D2	8	0	0	15	15	23
source-omr-31031-09	D1	8	2	0	8	8	8
source-omr-31031-10	D2	8	2	0	10	10	10
source-omr-31031-11	D2	8	4	0	14	14	14
source-omr-31031-12	D2	8	4	0	15	15	15
roadster-body-context-0-01	D4	9	1	0	34	32	954
roadster-body-context-1-01	D4	9	1	0	34	32	953
roadster-body-context-0-02	D4	9	1	0	34	30	1018
roadster-body-context-1-02	D4	9	1	0	34	30	1017
roadster-body-context-0-03	D4	9	1	0	34	30	847
roadster-body-context-1-03	D4	9	1	0	34	30	846
roadster-body-context-0-04	D4	9	1	0	32	30	414
roadster-body-context-1-04	D4	9	1	0	32	30	413